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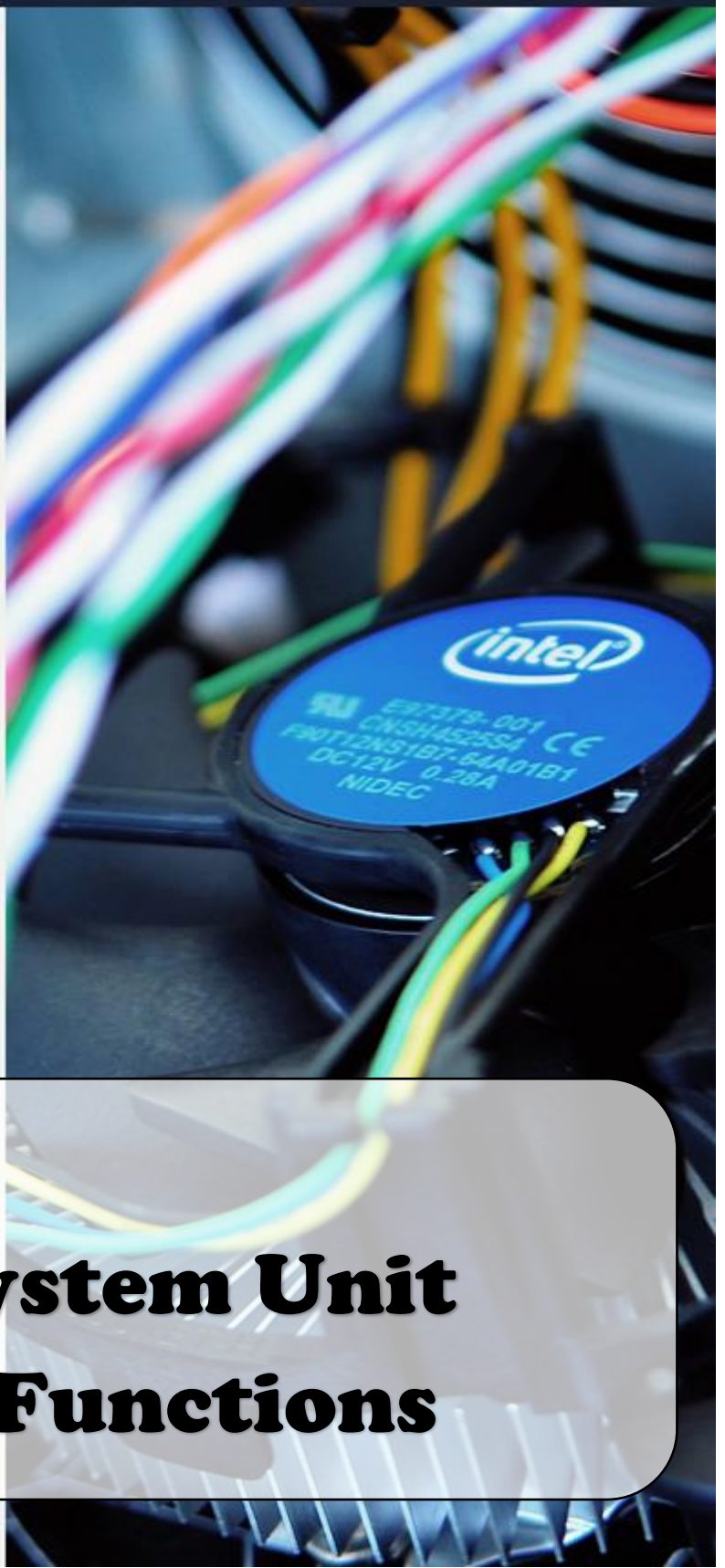
# COMPUTER SYSTEMS SERVICING NC II
# GRADE 11

## QUARTER 1
## COC 1:
## Install and Configure Computer Systems




## Module 3:
## Parts of System Unit and their Functions

**Computer Systems Servicing NCII – Grade 11**

**Quarter 1 – Module 3: Parts of System Unit and their Functions**

**First Edition, 2020**

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# Introductory Message

For the facilitator:

Welcome to the **Computer Systems Servicing NCII** module on **Parts of System Unit and their Functions**. This module was collaboratively designed, developed and reviewed by educators from Schools Division Office of Pasig City headed by its Officer-In-Charge Schools Division Superintendent, Ma. Evalou Concepcion A. Agustin in partnership with the Local Government of Pasig through its mayor, Honorable Vico Sotto.

The writers utilized the standards set by the K to 12 Curriculum using the Most Essential Learning Competencies (MELC) while overcoming their personal, social, and economic constraints in schooling.

This learning material hopes to engage the learners into guided and independent learning activities at their own pace and time. Further, this also aims to help learners acquire the needed 21st century skills especially the 5 Cs namely: Communication, Collaboration, Creativity, Critical Thinking and Character while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:

***Notes to the Teacher***

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Moreover, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to Computer Systems Servicing NCII Module on Parts of System Unit and their Functions. The hand is one of the most symbolized part of the human body. It is often used to depict skill, action and purpose. Through our hands we may learn, create and accomplish. Hence, the hand in this learning resource signifies that you as a learner is capable and empowered to successfully achieve the relevant competencies and skills at your own pace and time. Your academic success lies in your own hands!

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning material while being an active learner.

This module has the following parts and corresponding icons:

**Expectation** - These are what you will be able to know after completing the lessons in the module

**Pre-test -** This will measure your prior knowledge and the concepts to be mastered throughout the lesson.

**Recap** - This section will measure what learnings and skills that you understand from the previous lesson.

**Lesson-** This section will discuss the topic for this module.

**Activities** - This is a set of activities you will perform.

**Wrap Up**- This section summarizes the concepts and applications of the lessons.

**Valuing**-this part will check the integration of values in the learning competency.

**Post-test** - This will measure how much you have learned from the entire module. Ito po ang parts ng module

# EXPECTATION

After completing this lesson, you should be able to:

1. identify the parts of the system unit;
2. understand functions of each component of system unit;
3. appreciate the importance of identifying and understanding the components of System Unit.

# PRE-TEST

**DIRECTION:** Identify the following components of System Unit. Write your answer on the space provided before the number.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

## RECAP

**DIRECTION:** Identity the following statement carefully. Write **TRUE** if the statement is correct and **FALSE** if it's not. Write your answer on the space provided before the number.

_____1. The monitor is an output device used to display information.
_____2. A keyboard is an input device that allows a person to enter letters, numbers, and other symbols
_____3. A computer mouse is a handheld hardware output device that controls a cursor in a GUI (graphical user interface) and can move and select text, icons, files, and folders on your computer.
_____4. A VGA cable is use to connect the monitor into system unit to display data.
_____5. The System Unit is a part of computer system which encloses all the internal components of a computer.

# LESSON

The system unit is the box-like case that contains the electronic components of a computer including the motherboard, CPU, RAM, and other components. The system unit also includes the case that houses the internal components of the computer. Many people erroneously refer to this as the CPU.

The following are the components of system unit.

**1. Motherboard**

- It is sometimes called the *system board or main board*.
- It is the main circuit board of a microcomputer.
- This component contains the connectors for attaching additional boards.

**2. Power Supply Unit**

- This is the device that supplies power to your personal computer.
- It regulates the voltage to eliminate spikes and surges common in most electrical systems.

**3. Hard Disk Drive**

- This is the computer's main storage device used to store all data on the computer permanently.

- There are two types of HDD according to its connector:
  1. SATA (Serial ATA)
  2. EIDE (Enhanced Integrated Drive Electronics)

**4. Optical Disk Drive**

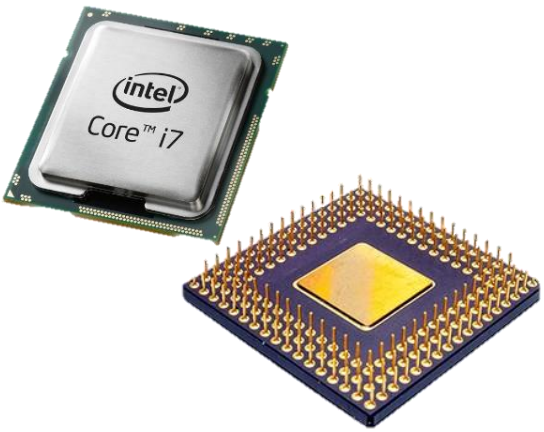
- This type of drive allows a user to retrieve, edit and delete the content from optical disks such as CDs, DVDs and Blu-ray disks.
- There are two types of ODD according to its connector:
    1. SATA (Serial ATA)
    2. EIDE (Enhanced Integrated Drive Electronics)

**5. RAM (Random Access Memory)**

- It is the memory which stores data until the machine is working.
- It allows stored data to be accessed randomly.
- Its main function is to store the data temporarily.

**6. CPU (Central Processing Unit)**

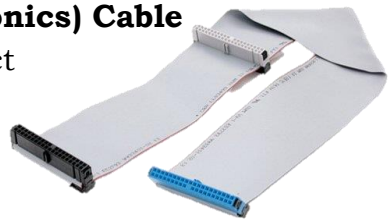
- It is the brain of the computer.
- Responsible for processing arithmetic & logical operations
- It runs the operating system and applications, constantly receiving input from the user or active software programs.




**7. CPU Cooling System**

- Also known as CPU Fan & Heatsink
- It is designed to reduce the ambient temperature from the CPU to keep the computer in running condition.

**8. EIDE (Enhanced Integrated Drive Electronics) Cable**



- It is a standard interface used to connect EIDE type of Drives such as Hard Disk Drive (HDD), CD-ROM or DVD-ROM.

**9. SATA (Serial ATA) Cable**

- It is used to connect SATA type Drives such as HDD, Optical Drives and Solid State Drives (SSD).

**10. Video Card**

- It is also known as graphics card
- It is use to generate and display the output images to a computer monitor.
- Can be connected to monitor using The following data cable:
  1. VGA Cable
  2. HDMI Cable
  3. DVI Cable

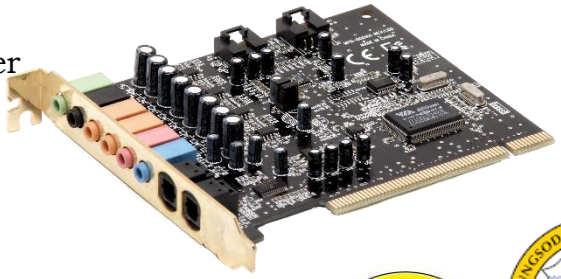
**11. NIC (Network Interface Card)**

- Also known as LAN Card or network adapter card
- It is use connect computers to a network using network cables.

**12. Wireless NIC (Network Interface Card)**

- It is use connect computers to a network using (radio frequency)
- Capable of establishing wireless network

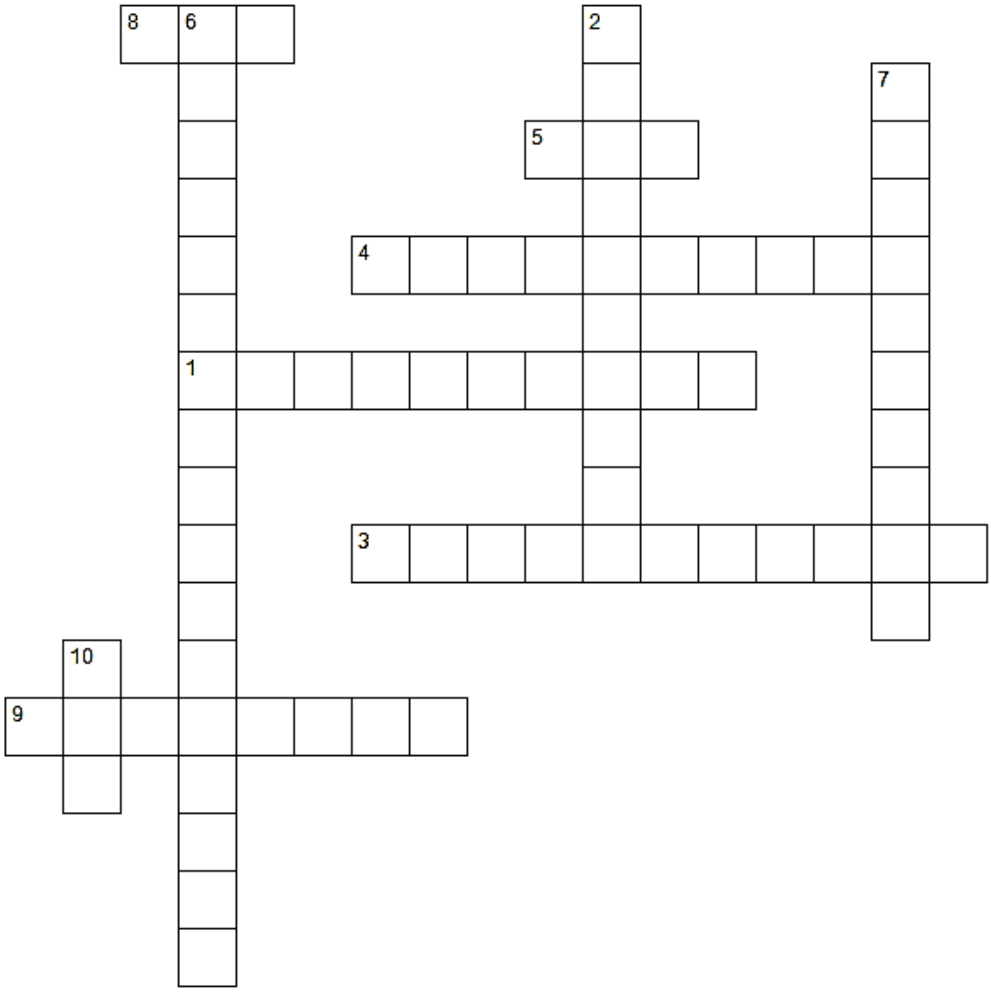
**13. Sound Card**



- It is also known as Audio Card Adapter
- It provides input and output of audio signals to and from a computer under control of computer programs.

# ACTIVITIES

**I. DIRECTION:** Complete the crossword by filling in a word that fits each clue.




Across:

1. It is also known as Audio Card Adapter.
3. It is the main circuit board of a microcomputer.
4. It is used to connect SATA type Drives (HDD/ODD) to the motherboard.
5. This component reads and writes data from optical disks through laser beaming technology.
8. It is considered as the brain of the computer.
9. This device allows computer to be connected in a network using cables.

Down:

2. It connects EIDE drives to the motherboard.
6. This is the device that supplies power to your personal computer.
7. It is use to generate and display the output images to a computer monitor.
10. It allows stored data to be accessed randomly.

II. **DIRECTION**: Read each statement below carefully. Write TRUE if the statement is correct, otherwise FALSE if it's not. Write your answer on the space provided.

_____1. The Voltage Regulator regulates the voltage to eliminate spikes and surges common in most electrical systems.
_____2. Video Card provides input and output of audio signals to and from a computer under control of computer programs.
_____3. Random Access Memory runs the operating system and applications, constantly receiving input from the user or active software programs.
_____4. Optical Disk Drive type of drive that allows a user to retrieve, edit and delete the content from optical disks such as CDs, DVDs and Blu-ray disks.
_____5. Hard Disk Drive is the computer's main storage device used to store all data on the computer permanently.

## W R A P – U P

**DIRECTION**: Complete the following sentences base on what you've learn.

1. System Unit is the _____
_____
_____
_____.

2. The parts of the System Unit are as follows _____
_____
_____
_____.

3. Write the functions of the following components of System Unit,
   a. Motheboard _____
   _____.
   a. AVR _____
   _____.
   a. CPU _____
   _____.
   a. RAM _____
   _____.

## VALUING

**DIRECTION:** Read the following questions carefully. State your answer in two to three sentences each number on the space provided.

1. Why is it important for us to learn the components of System Unit?

   _____
   _____
   _____
   _____

2. How will you use the knowledge you acquired about components of System Unit?

   _____
   _____
   _____
   _____

3. Cite a situation in which you can apply the knowledge of understanding components of System Unit.

   _____
   _____
   _____
   _____

## POST TEST

**DIRECTION:** Identify the following components of system unit. Write your answer on the space provided.

_____1. It is used to connect SATA type Drives (HDD/ODD) to the motherboard.

_____2. This device is Capable of establishing wireless network.

_____3. It is also known as graphics card and use to generate and display the output images/information to a computer monitor.

_____4. This component contains the connectors for attaching additional boards.

_____5. It runs the operating system and applications, constantly receiving input from the user or active software programs.

_____6. It is used to connect SATA type Drives (HDD/ODD) to the motherboard.

_____7. It is designed to reduce the ambient temperature from the CPU to keep the computer in running condition.

_____8. It provides input and output of audio signals to and from a computer under control of computer programs.

_____9. This type of drive allows a user to retrieve, edit and delete the content from optical disks such as CDs, DVDs and Blu-ray disks.

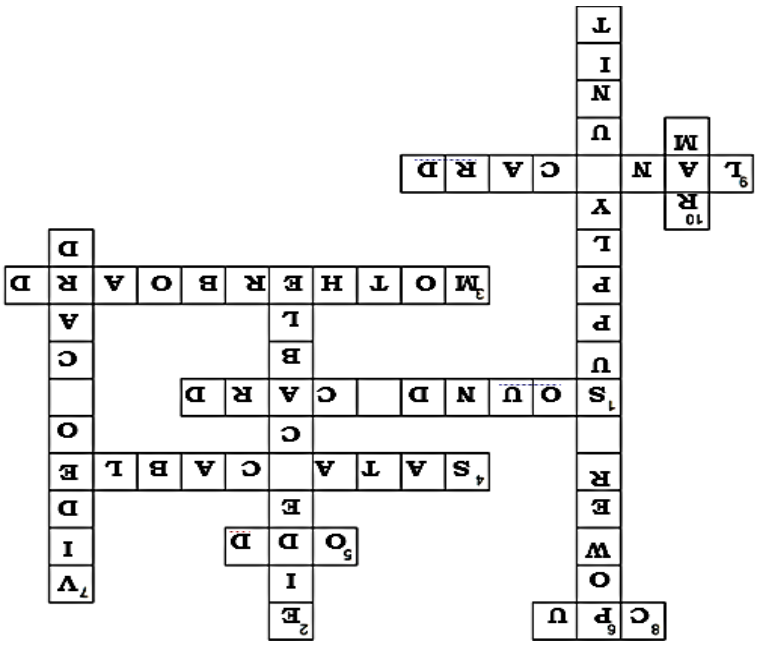
_____10. Its main function is to store the data temporarily.

# KEY TO CORRECTION

| PRE TEST | RECAP | POST TEST |
|---|---|---|
| 1. Video Card<br>2. Motherboard<br>3. (HDD) Hard Disk Drive<br>4. (RAM)Random Access Memory<br>5. ODD (Optical Disc Drive)<br>6. PSU (Power Supply Unit)<br>7. SATA Cable<br>8. EIDE Cable<br>9. CPU (Central Processing Unit)<br>10. CPU Cooling System | 1. True<br>2. True<br>3. False<br>4. True<br>5. True | 1. SATA Cable<br>2. Wireless NIC<br>3. Video Card<br>4. Motherboard<br>5. Central Processing Unit<br>6. 6SATA Cable<br>7. CPU Cooling System<br>8. Sound Card<br>9. Optical Disc Drive<br>10. Random Access Memory |

**ACTIVITY**

**I.**



**II.**

1. TRUE
2. TRUE
3. FALSE
4. FALSE
5. TRUE

# R E F E R E N C E S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